

RAGHAV

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LinkedIn

GitHub

LeetCode

Education

National Institute of Technology, Delhi

B.Tech in Electrical Engineering (CGPA: 7.46)

2024 – 2028

New Delhi, India

Maharshi Patanjali Vidya Mandir

Class XII – 93%

Class X – 93.6%

Prayagraj, India

2024

2022

Technical Skills

Languages: C, C++, JavaScript, Python

Web Development: HTML, CSS, MERN Stack (MongoDB, Express.js, React, Node.js), REST APIs, JWT, Zod, Multer, PDF-Parser

Machine Learning & AI: PyTorch, LangChain, LangGraph, FAISS, HuggingFace Transformers, HuggingFace Embeddings, RAG, Q-Learning

Core CS: DSA, OOP, OS, Computer Networks, DBMS

Tools: Git, GitHub, Postman, Streamlit, FAISS, MySQL, Razorpay API

Experience

Motilal Nehru National Institute of Technology (MNNIT) Allahabad

06/2026 – 07/2026

Research Intern

Prayagraj, India

- Multi-UAV Cooperative Search under GPS-Denial (MARL) | Python, NumPy – Built a Dec-POMDP environment simulating GPS-denied UAV search using LoRaWAN ToA ranging and battery-aware return-to-base logic.
- Ran realistic-vs-ideal-GPS comparison (12K episodes/condition); realistic constraints cut coverage/target-finding ~27% while keeping safe-return rate at 98%+.

Projects

AI Resume Analyzer & ATS Optimization Platform | [Live Demo](#) | [GitHub](#)

React.js, Tailwind CSS, Node.js,

Express.js, MongoDB, Google Gemini API

- Built a full-stack AI-powered resume analyzer with secure **JWT authentication**, PDF parsing, and resume version management.
- Integrated **Google Gemini with Zod validation** to generate structured ATS scores, keyword analysis, and AI-powered resume improvements.
- Developed analytics dashboards for tracking ATS trends and optimized MongoDB queries for high-efficiency data retrieval.

CodeCompass – AI Codebase Understanding Platform | [Live Demo](#) | [GitHub](#)

Python, FastAPI, React.js,

LangChain, LangGraph, FAISS, BM25, Groq API, Render, Vercel

- Built a full-stack AI platform that ingests public GitHub repositories and enables natural-language question answering over codebases using a FastAPI backend, React frontend, and REST APIs.
- Implemented a **hybrid RAG pipeline** combining FAISS vector search and BM25 keyword retrieval, with LangGraph-based conditional retrieval routing, multi-turn conversation memory, and confidence-scored source retrieval.
- Developed language-aware code chunking based on function and class boundaries to improve retrieval quality, and deployed the backend on Render and frontend on Vercel as independently scalable services.

ShopSmartAI – AI-Powered E-Commerce Platform | [Live Demo](#) | [GitHub](#)

React.js, Node.js, Express.js, MongoDB,

Redux Toolkit, Tailwind CSS, Cloudinary, Razorpay API, Python (Flask), Groq Llama 3, Surprise (SVD)

- Developed a full-stack AI-powered E-Commerce platform with secure authentication, product management, shopping cart, Razorpay payment integration, order processing, and a role-based admin dashboard.
- Built **ShopSmartAI** using Groq Llama 3 to enable natural-language product search, AI-powered product comparison, and intelligent shopping assistance.
- Designed a collaborative filtering recommendation engine using **SVD matrix factorization (Surprise)**, deployed as a Python Flask microservice integrated with the Express backend to deliver personalized recommendations for 50+ products across 200+ users.

Positions of Responsibility

- Training & Placement Cell** – Calling Coordinator
- Google Developers Club (GDC)** – Executive Member

Achievements

- Solved 800+ DSA problems** across LeetCode, GeeksForGeeks, and Coding Ninjas with consistent focus on time & space optimization.
- Ranked in **Top 1% in JEE Mains 2024** out of 15 lakh+ candidates nationwide.